§5 Induction

Instead of writing out all the steps (that we took in the rading check in #4), we'll agree there is a shortcut

New Rule

$P(1)$ (Base Case)

$\forall n \geq 1 (P(n) \to P(n+1))$ (Inductive Step)

$\therefore \forall k \geq 1 P(k)$ (Conclusion)

The above is mathematical induction

We can now prove $P(12456789)$ easily

1. $P(1)$
2. $\forall n \geq 1 (P(n) \to P(n+1))$

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3. $\forall k \geq 1 P(k)$ (Mathematical Induction on 1. and 2.)

$\therefore P(12456789)$ (Universal instatiation of 3.)

Warning

Neither the base case nor the inductive step is inherently easy, or inherently difficult

Exmaple

Let $Q(n)$ be the propositional function "n people is enough to form a team in Aussie rules footbal"

Let k be a generic positive integer, assume $Q(k)$ is true, "k people can form a team"

Q: What can we do, if $k+1$ people show up?

A: Have one person sit in a corner, and use the fact that k people can form a team, to make a team out of whoever is left.

Under these assumptions we see $Q(k+1)$ must be true.

Since we made an assumption, instead of proving $Q(k+1)$ (Unconditionally)

We've proved $Q(k) \to Q(k+1)$

Since k was generic (we used no k-facts)

We can apply a universal generalization, and conclude:

$\forall n \geq 1 Q(n) \to Q(n+1)$ (Inductive Step)

We don't yet have a base case.

Point 2

The base case and ionductive step can be dealth with in any order.

If we learn that at least once a team fit on a standard 72 person bus,, we can conclude $Q(72)$

It seems we should be able to conclude $Q(72)$ & $Q(1000)$ & $Q(10000000)$ etc.

Let $R(n)$ be $Q(n+71)$

So we know

$R(1)$ is true (Because $Q(72)$ is true)

but also if $n \geq 1$ then,

$R(n) \equiv Q(n+71)$ and

$Q(n+71) \rightarrow Q(n+1+71)$ and

$Q(n+1+71) \equiv R(n+1)$ thus

$\forall n \geq 1 (R(n) \rightarrow R(n+1))$

We can apply induction directly to R and conclude:

$\forall k \geq 1 R(k)$

but $R(k) \equiv Q(k+71)$

so if $n \geq 72$, then $Q(n) \equiv R(N-71)$ (which is true)

and thus $\forall k \geq 72 Q(k)$